Monte Carlo Integration

A Technique for Numerical Integration Using Random Numbers

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1. ABSTRACT

Monte Carlo integration is a numerical method that relies on random sampling to approximate the value of a definite integral. While other methods exist to perform numerical integration that leverage deterministic algorithms, Monte Carlo relies on a non-deterministic approach that yields a new outcome every time it is performed. It is upon averaging many outcomes from this method that, due to the Law of Large Numbers, the results converge on the true value of the definite integral. Increasing the number of samples reduces the error. A probabilistic background will be developed to provide a foundation for the method, while other important considerations are explored, as well. A tutorial on how to perform Monte Carlo integration will be presented to support the idea, while providing examples of the usefulness of this technique.

2. BACKGROUND

2.1 Random Variables

To understand Monte Carlo integration, it is important to build up a foundation to grasp why the technique works so well. A random variable, say X, is a variable whose numerical value is determined by the outcome of a random phenomenon, like flipping a coin and generating heads or tails, or the temperature outside tomorrow [1]. These variables can also be discrete, where there is a fixed set of outcomes from the random phenomenon like heads or tails, X = {H, T}, or they can be continuous, where the result can take real values in R, like the temperature outside tomorrow [1]. The following equations and information relevant to the mathematical foundation for Monte Carlo integration were adapted from [1] and will be noted as such.

2.2 Cumulative Distribution and Density Functions

The cumulative distribution function, CDF, calculates the probability that a random variable X will take on a value less than or equal to a specified value x [1]:

$$cdf(x) = P\{X \leq x\}$$

The related probability density function, PDF, is the derivative of the CDF [1]:

$$pdf(x) = \frac{d}{dx} cdf(x)$$

The CDF is always increasing, which means that the PDF is always non-negative, and a very important relationship is developed from the two previous equations, allowing us to compute the probability that a random variable lies within an interval [1]:

$$P\{a \leq X \leq b\} = \int_a^b pdf(x)\, dx$$

Given this previous expression, it is apparent that the PDF must always integrate to one over the full expanse of its domain [1].

2.3 Expected Value and Variance

The expected value of a random variable Y = f(x) over a domain μ(x) is defined as [1]:

$$E[Y] = \int_{\mu(x)} f(\mathrm{x})\mathrm{pdf}(\mathrm{x})d\mu(x).$$

With the variance being [1]:

$$\sigma^2[Y] = E[(Y - E[Y])^2]$$

Where σ, the standard deviation, is the square root of the variance [1]. With these definitions, it is easy to show that for any constant a [1],

$$E[aY] = aE[Y],$$

$$\sigma^2[aY] = a^2\sigma^2[Y].$$

The expected value of a sum of these random variables Yi is the sum of their expected values [1]:

$$E\left[\sum_i Yi\right] = \sum_i E[Yi].$$

From the properties outlined above, it is possible to derive a simpler expression for the variance [1]:

$$\sigma^2[Y] = E[Y^2] - E[Y]^2.$$

And if the random variables are uncorrelated and independent, a summation property can be developed for the variance [1]:

$$\sigma^2\left[\sum_i Yi\right] = \sum_i \sigma^2[Yi].$$

2.4 The Monte Carlo Estimator

Monte Carlo integration leverages random sampling of a function to approximate the value of a definite integral [1]. Suppose we want to integrate the one-dimensional function f(x) from a to b [1]:

$$F = \int_a^b f(x)\,dx.$$

We can approximate the integral by averaging samples of the function $f(x)$ at uniform random points within the interval [1]. Given a set of N uniform random variables $Xi \in [a, b)$ with a corresponding pdf of 1 / (b – a), the Monte Carlo Estimator for computing F is [1]:

$$\langle F^N \rangle = (b-a)\frac{1}{N-1}\sum_{i=0}^{n} fi(Xi).$$

The random variable $Xi \in [a, b)$ can be constructed by transforming a random variable uniformly distributed between zero and one, $Ui \in [0, 1)$ [1]:

$$Xi = a + (b-a)Ui.$$

Using this construct, we can expand the estimator to [1]:

$$\langle F^N \rangle = (b-a)\frac{1}{N}\sum_{i=0}^{n} f(a + (b-a)Ui).$$

Since $\langle F^N \rangle$ is a function of Xi, it itself is a random variable. $\langle F^N \rangle$ approximates F using N samples [1]. Intuitively, the first Monte Carlo estimator developed computes the mean value of the function f(x) over the interval a to b and then multiplies this mean by the length of the interval (b – a) [1]. By moving (b – a) into the summation, the estimator id randomly evaluating different heights of the function and averaging a set of rectangular areas computed by multiplying the height by the interval length (b – a) [1].

2.5 Expected Value and Convergence

It can be shown that the expected value of $\langle F^N \rangle$ is in fact F [1]:

$$E[\langle F^N \rangle] = E\left[(b-a)\frac{1}{N}\sum_{i=0}^{N-1} f(Xi)\right],$$

$$= (b-a)\frac{1}{N}\sum_{i=0}^{N-1} E[f(Xi)],$$

$$= (b-a)\frac{1}{N}\sum_{i=0}^{N-1} \int_a^b f(x)pdf(x)dx,$$

$$= \frac{1}{N}\sum_{i=0}^{N-1} \int_a^b f(x)dx\,,$$

$$= \int_a^b f(x)dx,$$

$$E[\langle F^N \rangle] = F.$$

If we increase the number of samples N, the estimator $\langle F^N \rangle$ becomes an ever-closer approximation of F and due the Law of Large Numbers, in the limit we can guarantee that we have the solution [1]:

$$P\left\{\lim_{N\to\infty} \langle F^N \rangle = F\right\} = 1.$$

It is important to know how quickly this estimate converges to a sufficiently accurate solution and this can be analyzed by determining the convergence rate of the estimator's variance [1]. The standard deviation is proportional to [1]:

$$\sigma[\langle F^N \rangle] \propto \frac{1}{\sqrt{N}}.$$

Which means that we must quadruple the number of samples to reduce the error by half [1]. Standard integration techniques exist which converge much faster in one dimension; however, the techniques suffer from the curse of dimensionality, where the convergence rate grows exponentially worse as dimensions are increased [1]. The basic Monte Carlo estimator can be extended to multiple dimensions and the convergence rate for Monte Carlo $O(\sqrt{N})$ is independent of the number of dimensions in the integral, thus making Monte Carlo a very practical method for approximating integrals in higher dimensions [1].

2.6 Multidimensional Integration

Monte Carlo integration can be generalized to higher dimensions, as mentioned previously, by continuing to use random variables from PDFs to compute the integrals [1].

$$F = \int_{\mu(x)} f(x) d\mu(x),$$

And the following [1]:

$$E[\langle F^N \rangle] = \frac{1}{N} \sum_{i=0}^{N-1} \frac{f(Xi)}{pdf(Xi)} .$$

It can also be shown that the generalized estimator also has the correct expected value [1]:

$$E[\langle F^N \rangle] = E\left[\frac{1}{N} \sum_{i=0}^{N-1} \frac{f(Xi)}{pdf(Xi)}\right],$$

$$= \frac{1}{N} \sum_{i=0}^{N-1} E\left[\frac{f(Xi)}{pdf(Xi)}\right],$$

$$= \frac{1}{N} \sum_{i=0}^{N-1} \int_{\Omega} \frac{f(x)}{pdf(x)} \, pdf(x) dx ,$$

$$= \frac{1}{N} \sum_{i=0}^{N-1} \int_{\Omega} f(x) dx ,$$

$$= \int_{\Omega} f(x) dx ,$$

$$E[\langle F^N \rangle] = F.$$

In addition to the convergence rate, Monte Carlo integration also extends to multiple dimensions quite easily. Monte Carlo techniques provide the freedom of choosing any arbitrary number of samples [1].

3. TUTORIAL

Now, a step-by-step tutorial on how to perform Monte Carlo integration will be shown below with a simple breakdown of the complex ideas previously presented.

First, define the function and the bounds over which you would like to integrate that function. This can be written as:

$$F = \int_a^b f(x)dx \,, a \leq x \leq b \,.$$

Suppose we have U_1, U_2,…, U_n are independent and identically distributed Unif(0,1), which are random numbers from the continuous uniform distribution with a minimum 0 and maximum 1, and we define from the above equation [2]:

$$F_i \equiv (b-a)f(a+(b-a)Ui) \; for \; i = 1,2,\dots,n$$

We can use the sample average as the estimator for the value of the integral for the defined function, F [2]:

$$\overline{F_N} \equiv \frac{1}{N}\sum_{i=0}^{N} F_i = \frac{b-a}{N}\sum_{i=0}^{N} f(a+(b-a)U_i).$$

Via the Law of the Unconscious Statistician, it follows that [2]:

$$E[\overline{F_N}] = (b-a)E[f(a+(b-a)U_i)]$$

$$= (b-a)\int_R f(a+(b-a)u)f(u)du$$

(where f(u) is the Unif(0,1) pdf)

$$E[\overline{F_N}] = (b-a)\int_0^1 f(a+(b-a)u)du = F$$

Due to the law of Large Numbers, if an estimator is asymptotically unbiased and the variance goes to zero then everything is working well [2]:

$$\overline{F_N} \to F \; as \; N \to \infty$$

Given this understanding, we can generate a large amount of Unif(0,1) observations to plug into the original function and solve the equation:

$$\overline{F_N} = \frac{b-a}{N}\sum_{i=0}^{N} f(a+(b-a)U_i).$$

3.1 Single Dimension Monte Carlo Integration

Suppose we have a function that might take a little effort to solve, say

$$F = \int_0^1 \sin(\pi x)\, dx \,,$$

Let's take 4 ($N = 4$) Unif(0,1) observations, $U_1 = 0.59$, $U_2 = 0.32$, $U_3 = 0.71$, $U_4 = 0.89$. Since,

$$F_i = (b-a)f(a+(b-a)U_i) = f(U_i) = \sin(\pi U_i),$$

We then obtain,

$$\overline{F_N} = \frac{1}{N}\sum_{i=1}^{N} F_i = \frac{1}{4}\sum_{i=1}^{4}\sin(\pi U_i) = 0.733378$$

Considering the true value of the integral given the bounds is 0.636619, it is not the best guess, but thankfully when following the Law of Large Numbers and utilizing many more Unif(0,1) observations, we eventually get much closer to the true value of the integral.

If we implement this in Python, we can create a function to test different values of N, which means the amount of Unif(0,1) observations used, until we converge on the true result. The code can be found in the accompanying files, but the results can be seen below.

In this example, a function to perform the Monte Carlo integration on the equation was defined. This function was then performed on different amounts of observations of Unif(0,1) i.e. values of N and plotted to observe how the results converge on the true value of integral. The range of Unif(0,1) observations was 10 – 100,000 and the red line is a polynomial fit. From figure 1 below, as the number of Unif(0,1) observations increases, the accuracy of the estimate increases thus supporting the main idea of this technique:

$$\overline{F_N} \rightarrow F \text{ } as \text{ } N \rightarrow \infty \text{ .}$$

When looking at the plot in figure 1 below, it appears that result dances around the true value but doesn't quite converge on it. In figure 2, at 50,000 observations sampled there is good convergence on the true value and this result only increases in figure 3 with 100,000 observations sampled.

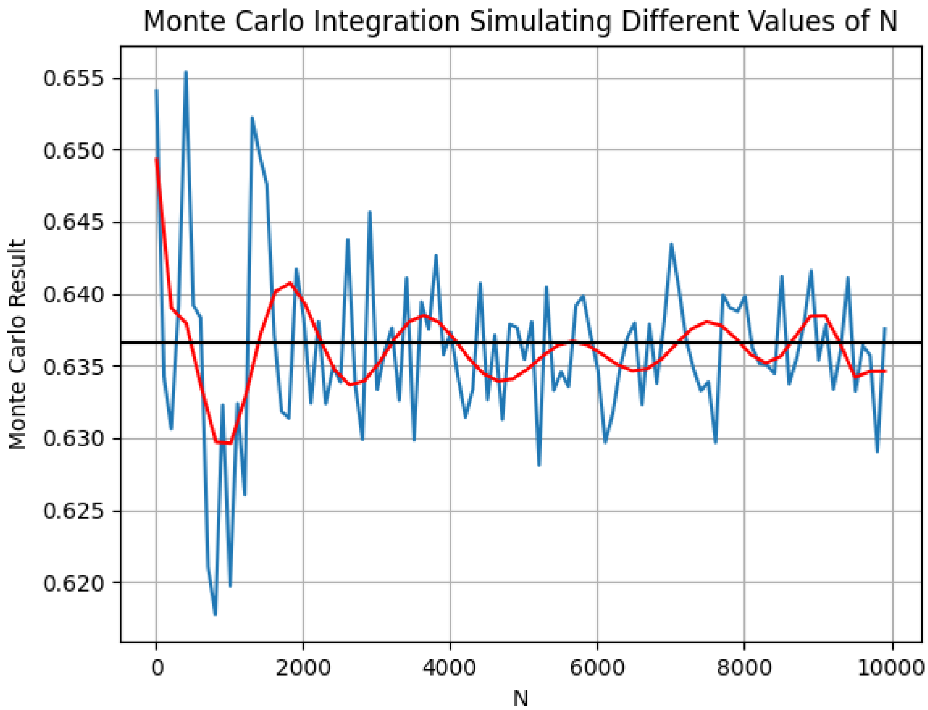

Figure 1: Monte Carlo integration result with respect to the number of Unif(0,1) observations (up to 10,000) sampled.

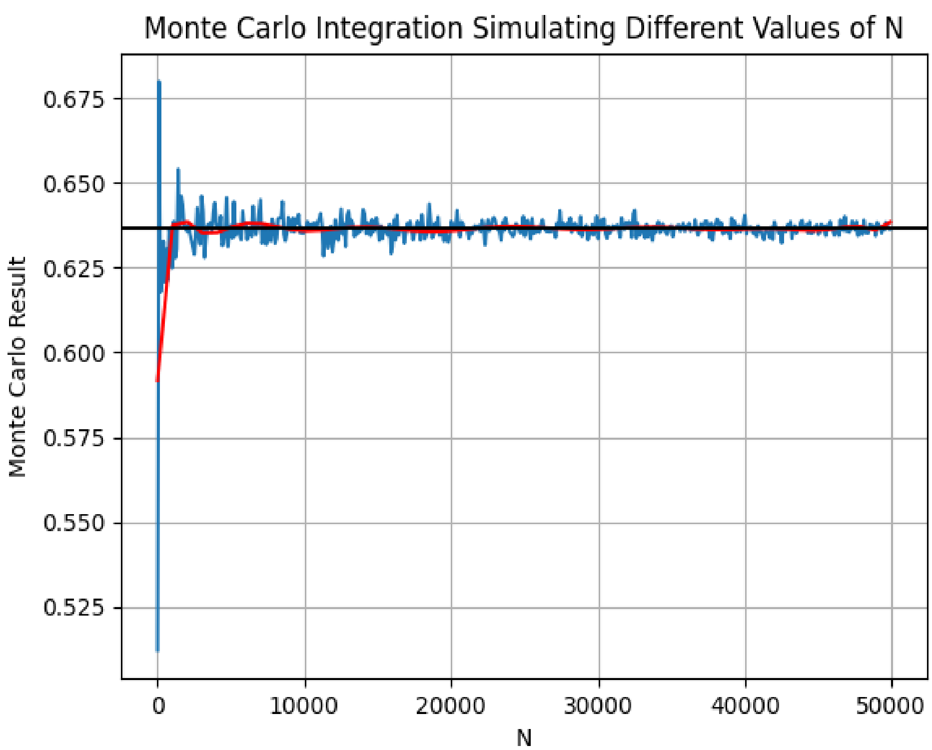

Figure 2: Monte Carlo integration result with respect to the number of Unif(0,1) observations (up to 50,000) sampled.

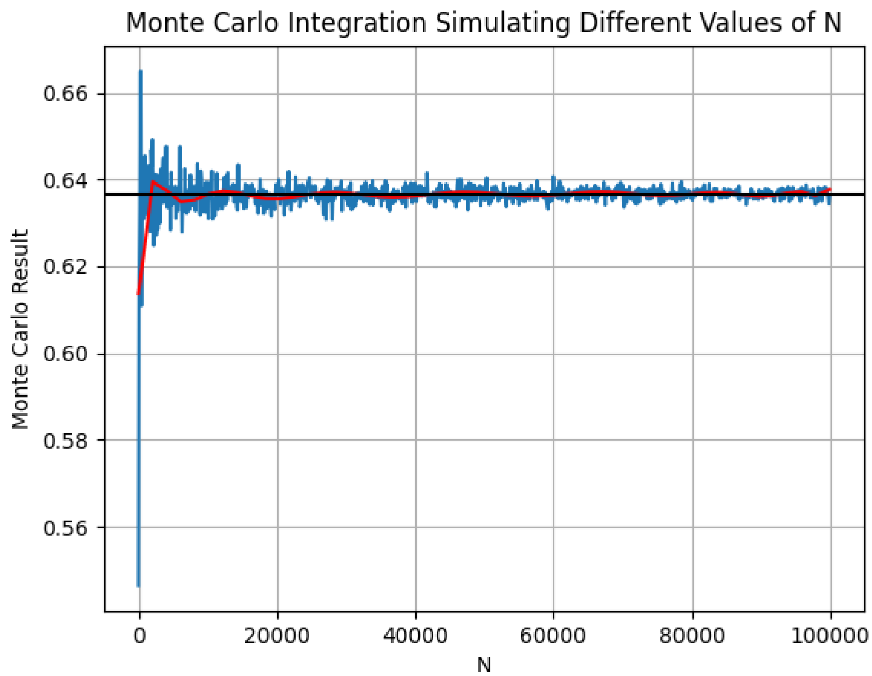

Figure 3: A plot of the Monte Carlo integration result with respect to the number of Unif(0,1) observations (up to 100,000) sampled.

However, upon closer examination, when plotting the relative error of the Monte Carlo integration estimation versus the true value, the relative error finally becomes considerably small around 100,000 observations sampled and it can be noted that the relative error converges at the same rate as $1/\sqrt{N}$ which is proportional to the standard deviation [3]. This phenomenon can be seen in figure 4 below.

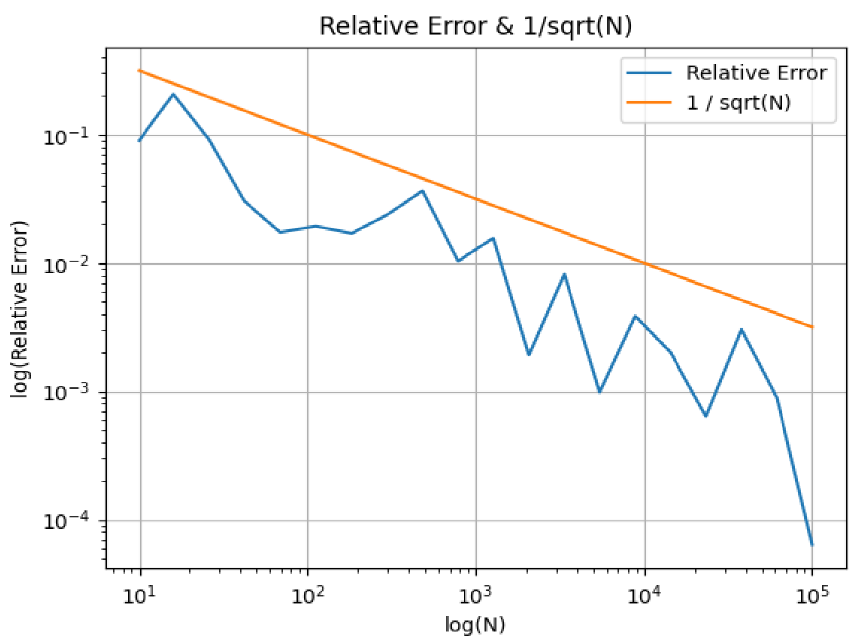

Figure 4: The relative error of the Monte Carlo integration converges at the same rate as $\frac{1}{\sqrt{N}}$. [3]

The sample variance, standard error, and the confidence interval for the 1D Monte Carlo integration, along with other relevant values can be found in the following table. This was completed with N = 100,000 Unif(0,1) observations. This is a good estimator since the true value of the integral lies within the bounds of the 95% confidence interval and there is low sample variance and standard error.

Table 1: Properties of 1D Monte Carlo Integration Simulation

Property	Value
N	100,000
True Value of the Integral	0.63661977
Monte Carlo Integral Estimation	0.63696265
Sample Variance	0.09481657
Standard Error	0.00097373
95% Confidence Interval	[0.63505412, 0.63887117]

3.2 Higher Dimension Monte Carlo Integration

Say we have a more complex and interesting function that we want to integrate:

$$f(x, y) = \sin(x^2) * \cos(y^2) \; 0 \le x \le 1, 0 \le y \le 1 \, .$$

$$F(x, y) = \iint_0^1 \sin(x^2) * \cos(y^2) \, dydx \; 0 \le x \le 1, 0 \le y \le 1 \, .$$

We can step through the same type of process to generate similar results as before in 1D. For this function, at 10,000 Unif(0,1) observations, we see it quickly begin to start oscillating around the true value of the definite integral. As we saw in the previous example with 1D space, the Monte Carlo estimator converges on the true value of the integral as the number of Unif(0,1) observations increases since it is asymptotically unbiased.

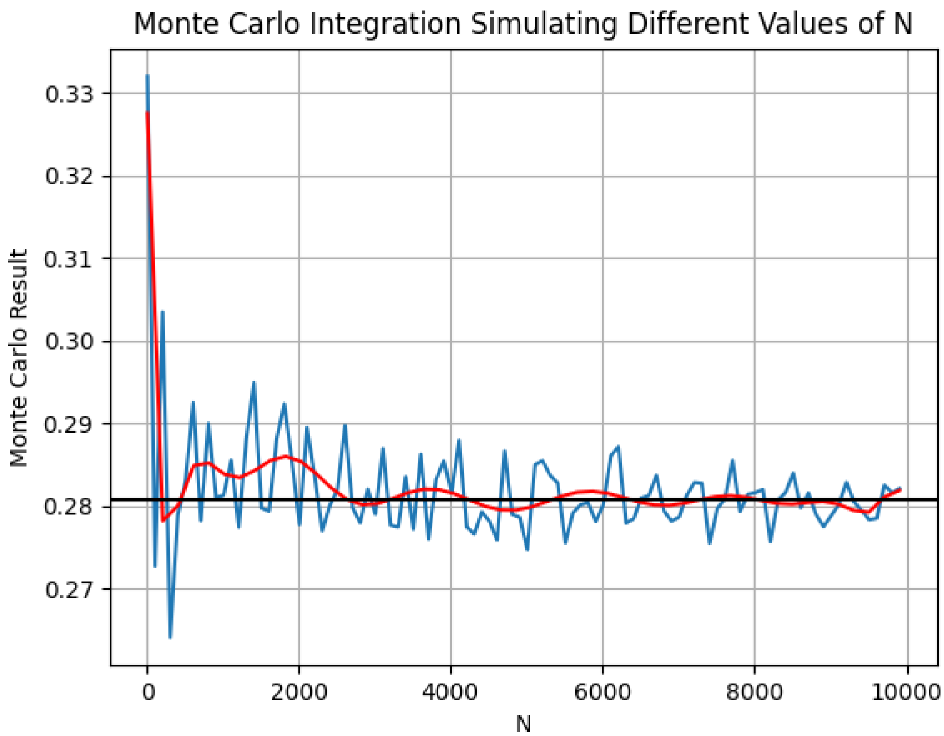

Figure 5: Monte Carlo integration result with respect to the number of Unif(0,1) observations (up to 10,000) sampled.

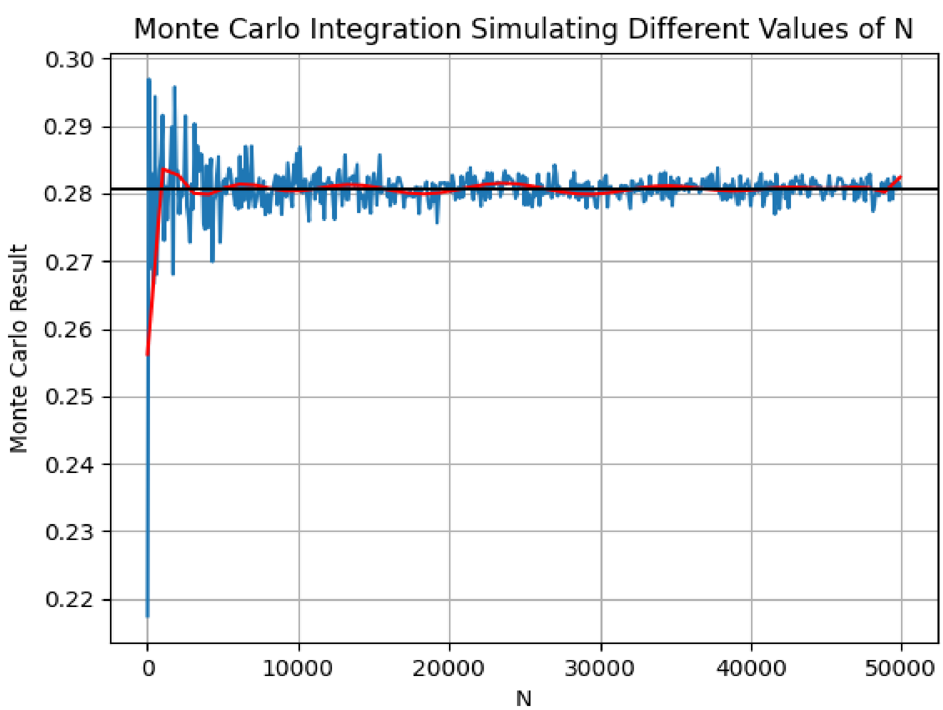

Figure 6: Monte Carlo integration result with respect to the number of Unif(0,1) observations (up to 50,000) sampled.

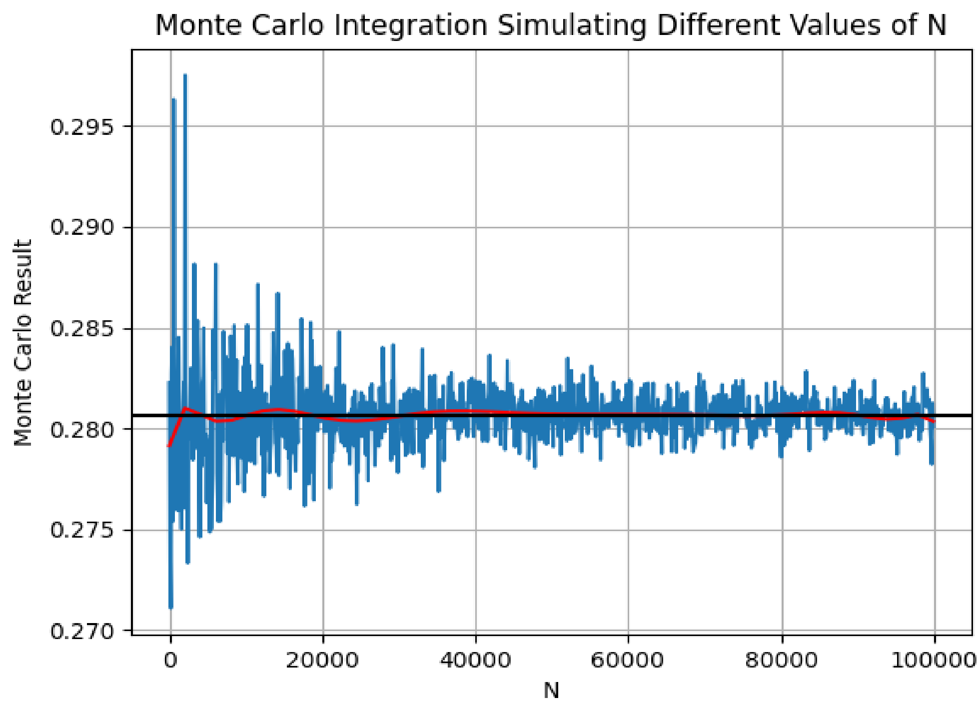

Figure 7: Monte Carlo integration result with respect to the number of Unif(0,1) observations (up to 100,000) sampled.

We can also observe that the same trend in relative error and the convergence rate holds true in 2D space, as well. This is a good confirmation that this method holds up for more than just 1D integration.

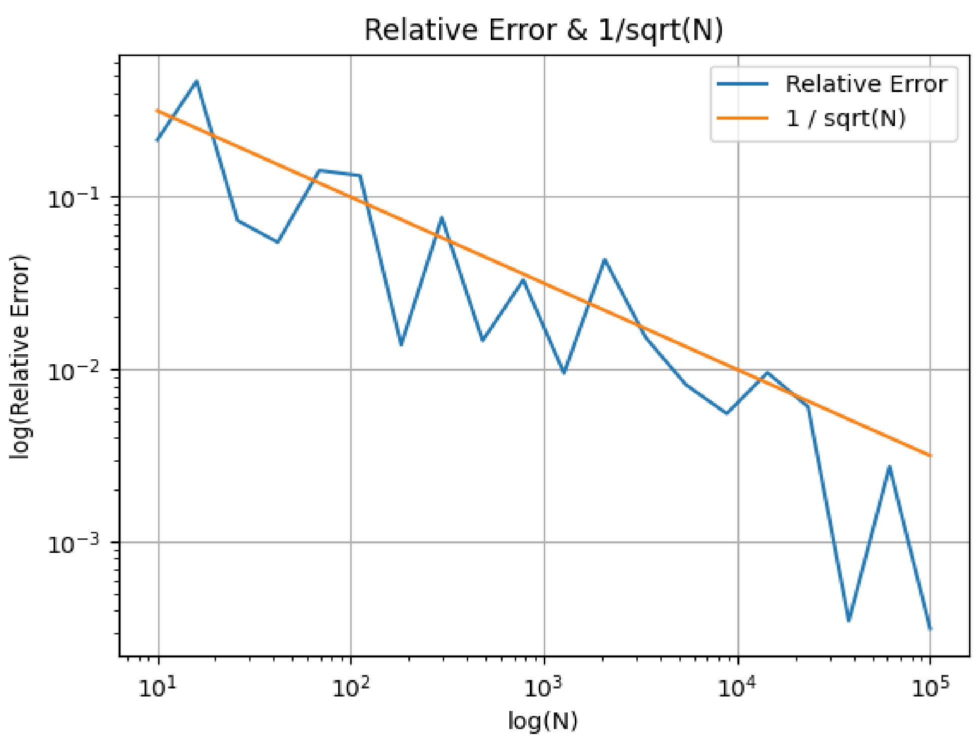

Figure 8: The relative error of the Monte Carlo integration converges at the same rate as $\frac{1}{\sqrt{N}}$. [3]

The sample variance, standard error, and the confidence interval for the 2D Monte Carlo integration, along with other relevant values can be found in the following table. This was completed with N = 100,000 Unif(0,1) observations. This is a good estimator since the true value of the integral lies within the bounds of the 95% confidence interval and there is low sample variance and standard error.

Table 2: Properties of 2D Monte Carlo Integration Simulation

Property	Value
N	100,000
True Value of the Integral	0.28064519
Monte Carlo Integral Estimation	0.28025228
Sample Variance	0.05957441
Standard Error	0.00077184
95% Confidence Interval	[0.27873946, 0.28176509]

This method easily extrapolates to higher dimensions as we have detailed in the background section of this paper and then most recently in the 2D space, where the Monte Carlo integration method continues to hold up and it converges on the true value of the function being evaluated.

4. Conclusion

From the results presented, it can be concluded that Monte Carlo integration is a useful technique for estimating the value of an integral. Due to its ability to scale to higher dimensions, since the convergence rate is independent of the number of dimensions in the integral. After looking at the ease of which it is to scale from single dimension to multiple dimensions, along with the speed of calculation for integrals of complex functions, it is no surprise that this is a very useful technique in many applications including machine learning where evaluating integrals with high dimensionality arises, computer graphics by estimating values of integrals from light sources or shapes of shadows, statistical mechanics where the study of phase change may involve complex integration, and so much more.

5. References

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